

AGRICULTURAL DECISION SUPPORT · ANDROID · CSE PROJECT

AgriSense

Intelligent Crop Decision & Risk Analysis System

“Understand your farm. Plan your crop. Reduce uncertainty.”

TEAM

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COLLEGE

GNIOT, Group of Institution

DEPARTMENT

Computer Science And Engineering

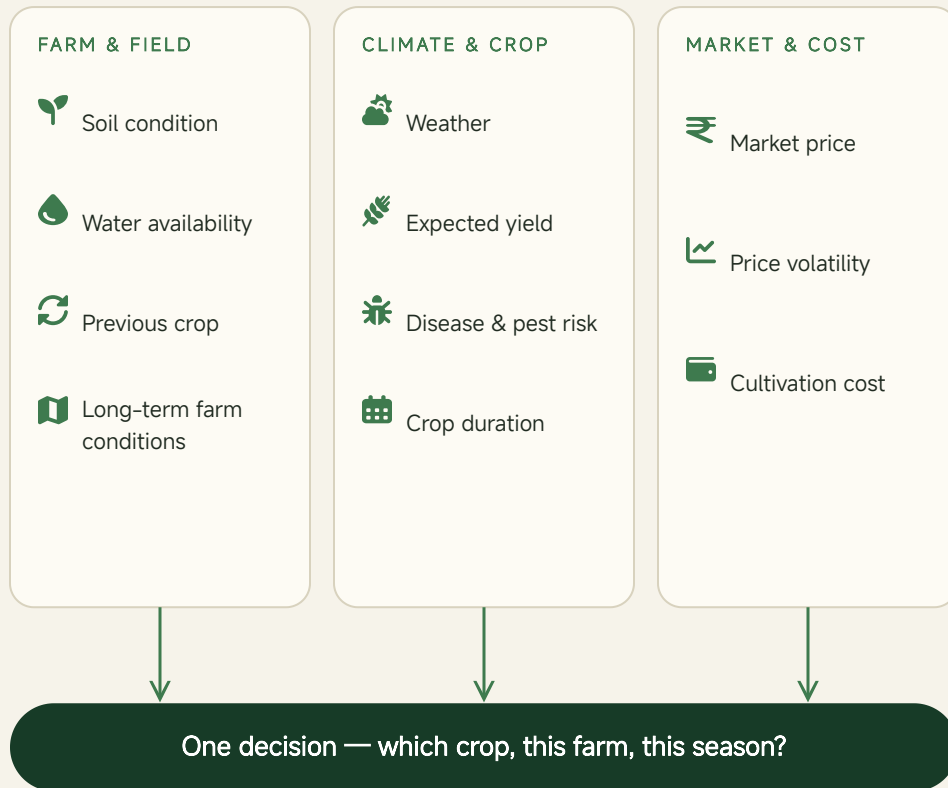
ACADEMIC YEAR

2026-27

THE PROBLEM

The farmer's decision is more complicated than choosing a crop

Choosing a crop means weighing field conditions, climate, biology and economics at the same time — eleven factors shape the outcome.



“A crop that looks profitable on paper may not be the best crop for a particular farm.”

Meet AgriSense

A farm-level decision-support app that evaluates crop options against farm conditions, environmental factors and economics — together, not in isolation.



Working prototype — combines farmer input with demonstration datasets and modeled estimates.

Not Just “What Should I Plant?”

SIMPLE RECOMMENDER

Soil looks suitable



Plant Crop X

One variable in, one answer out.

No economics, no uncertainty, no downside.

AGRISENSE

Crop X fits your soil — but what is the **expected profit**, the **water requirement**, the **risk**, the **crop duration** — and how does it behave in **downside scenarios**?

CORE CAPABILITIES



Crop suitability



Profit estimation



ROI



Risk analysis



Water requirement



Crop comparison



What-if simulation



Long-term planning

The app helps the farmer understand the decision — not blindly follow one recommendation.

From Farm Data to a Crop Decision

INPUTS



Soil data



Weather data



Water availability



Crop database



Market & cost data



DECISION CORE

Recommendation engine



Estimation

yield · revenue · profit



Risk analysis



What-if simulation



OUTPUT



Farmer Decision Report

ranked recommendations

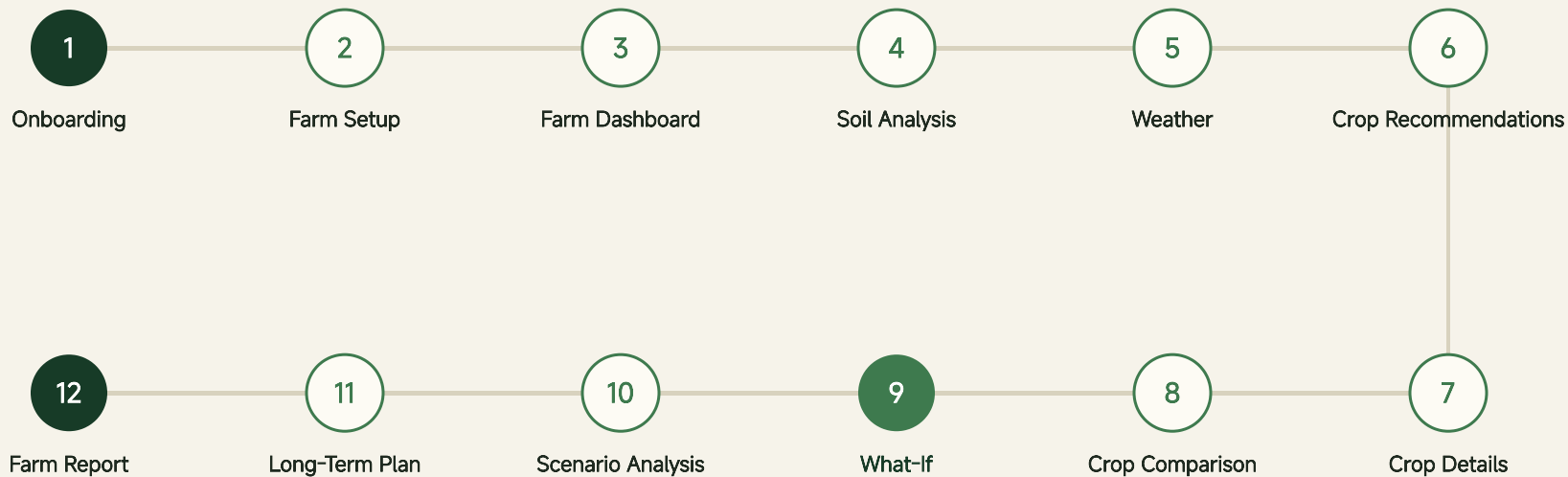
economics · risk

scenario comparison

Every stage runs on-device in the current prototype; data sources are demo datasets.

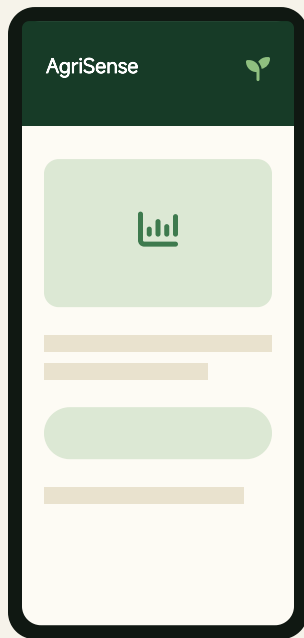
Inside the AgriSense App

Twelve screens, one continuous decision journey — from farm setup to a shareable report.

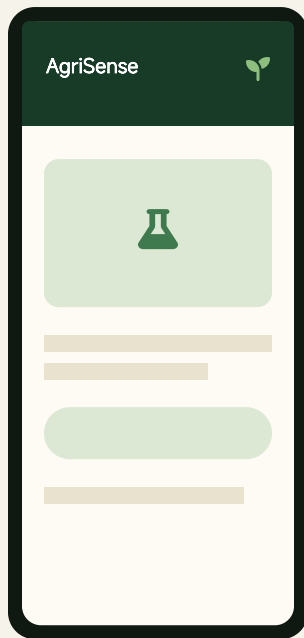


Each screen answers one question in the decision flow — no dead ends, no orphaned features.

AgriSense — Mobile Experience



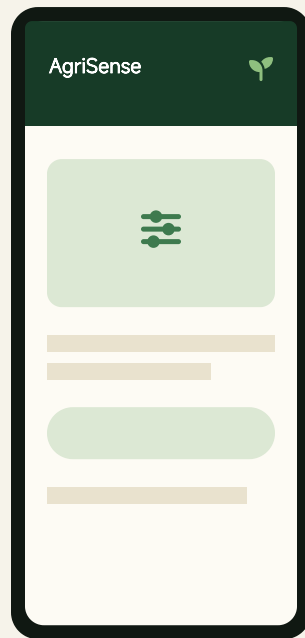
Dashboard



Soil Analysis



Crop Recommendations



What-If Simulator

Placeholder frames — app screenshots drop in at the same size, undistorted.

More Than a Single Recommendation

PROTOTYPE · DEMO ESTIMATES

CROP	SUITABILITY / 100	EXPECTED PROFIT (₹ / SEASON)	RISK	CONFIDENCE
 Maize 95–110 days Water: medium	87	₹43K – ₹68K	Medium	76%
 Mustard 110–140 days Water: low	82	₹38K – ₹61K	Low–Med	81%
 Chickpea 100–120 days Water: low	78	₹34K – ₹57K	Low	72%
 Rice 130–150 days Water: high	61	₹30K – ₹64K	High	65%

Modeled estimates from the prototype's demo dataset — illustrative comparison, not a live prediction.

What If?

ADJUST THE CONDITIONS

Rainfall



Temperature



Market price



Fertilizer cost



Water availability



MODELED SCENARIO OUTPUTS

Normal conditions

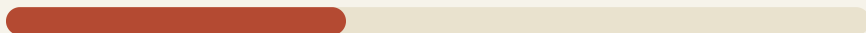
₹55K expected profit



Risk: medium

30% lower rainfall

₹28K expected profit



Risk: high

Higher market price

₹71K expected profit



Risk: medium

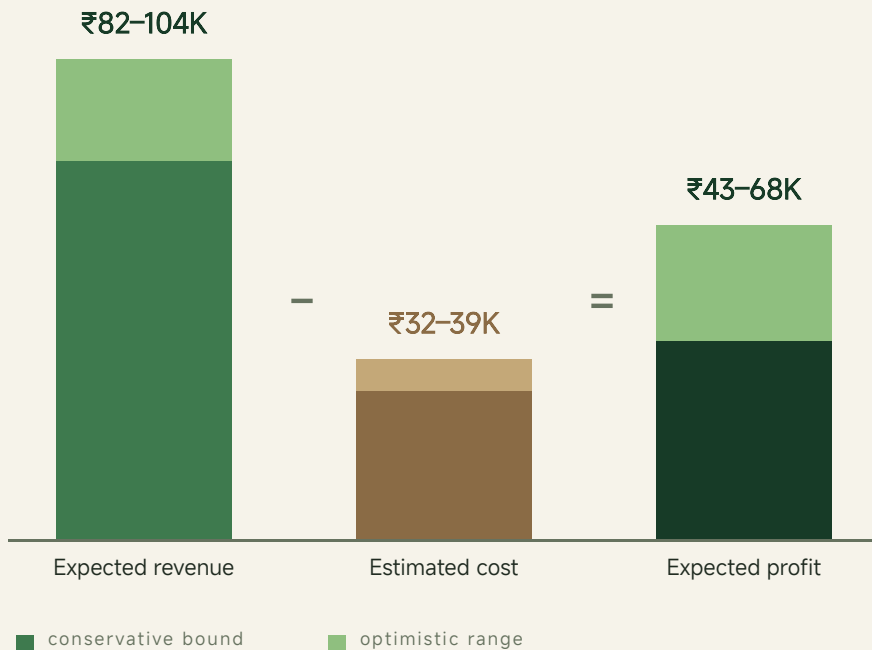
Scenario outputs are modeled demo values.

The farmer can explore how a decision behaves when conditions change — before committing seed, money and time.

From Yield to Profit

Expected revenue = expected yield × expected price

Expected profit = expected revenue – cultivation cost



ALSO COMPUTED PER CROP



ROI

Return on cultivation cost



Crop duration

Days from sowing to harvest



Profit per day

Profit normalised by crop duration



Market risk

Price volatility expected at harvest

Demo estimates from the prototype's sample crop dataset.

Profit Is Not the Only Objective

SIX RISK DIMENSIONS, SCORED PER CROP



Weather risk

Rainfall and temperature swings during the season



Water risk

Irrigation availability vs. crop water requirement



Market risk

Price volatility between sowing and harvest



Disease risk

Pest and disease pressure for the region and season



Soil risk

Nutrient depletion and degradation across seasons



Cost risk

Input price inflation — seed, fertilizer, labour

LONG-TERM CROP PLANNING

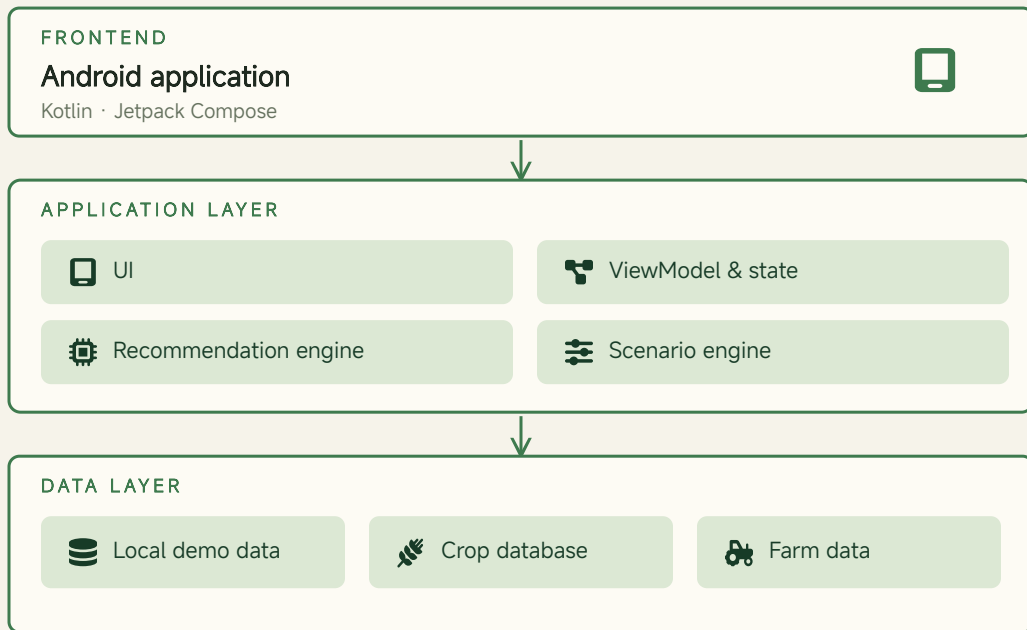


Rotation-aware: legumes restore soil nitrogen after cereal seasons.

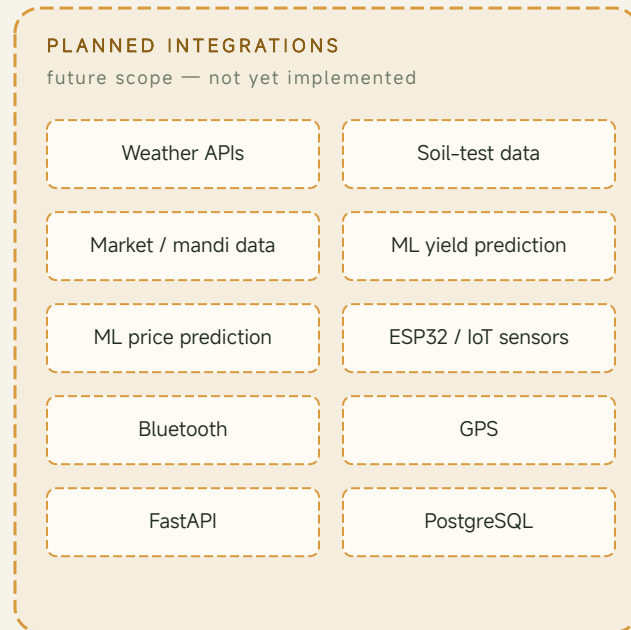
The system evaluates short-term economics while also weighing multi-year crop planning and rotation.

This year's profit should not become next year's problem — soil health and risk compound across seasons.

Technology Behind AgriSense



— implemented in prototype



— planned / future integration

The prototype runs fully on-device with local demo data; live sources connect through a FastAPI + PostgreSQL backend as they are added.

Where AgriSense Goes Next

CURRENT LIMITATIONS — PROTOTYPE

- Some modules run on demonstration / local data
- Real agricultural predictions require validated regional datasets
- Market and weather features need live API integration
- ML models require sufficient historical regional data
- Sensor measurements require calibrated hardware
- **Recommendations are decision support — not guaranteed outcomes**

DEVELOPMENT ROADMAP



Each phase is independent — the app stays useful at every stage of the roadmap, and each addition strengthens the decision engine.

CONCLUSION

AgriSense — Smarter Decisions, Not Blind Predictions



→ better-informed planting decisions, compared before sowing

“The goal is not to tell farmers what to do.

The goal is to help them make better-informed decisions.”

Thank You

Questions?